

5. Test Environment Setup

The **Test Environment Setup** section explains how the testing environment is prepared to execute API tests for the ReqRes REST APIs using Postman and Newman. A properly configured environment ensures smooth execution of test cases, reduces errors, and improves efficiency during API testing.

This setup includes installing required tools, configuring Postman environment variables, and preparing Newman for automation execution.

5.1 Postman Installation

Overview

Postman is the primary tool used for API testing in this project. It must be installed and configured before starting any testing activities.

Installation Steps

1. Open the official Postman website: <https://www.postman.com/downloads>
2. Download the version suitable for your operating system (Windows/Mac/Linux).
3. Run the installer file.
4. Complete the installation process by following on-screen instructions.
5. Launch Postman after installation.

Post-Installation Setup

- Create a free Postman account (optional but recommended).
- Login to enable cloud sync (collections, environments).
- Create a workspace for organizing API projects.

Importance

Postman provides a user-friendly interface to:

- Send API requests
- View responses
- Write test scripts
- Organize APIs into collections

It is the foundation tool for manual API testing in this project.

5.2 Environment Configuration

Overview

In Postman, an **Environment** is used to store and manage variables that can be reused across multiple API requests. This makes testing flexible and reduces repetition.

Environment Used in This Project

A single environment is created:

ReqRes API Environment

Key Variable Configured

Variable Name	Value
base_url	https://reqres.in/api

How It Works

Instead of writing full URLs in every request, we use:

```
{{base_url}}/users
```

This helps in:

- Easy maintenance
- Quick updates if URL changes
- Reusability across APIs

Steps to Create Environment

1. Open Postman
2. Go to Environments → Create New
3. Add variable `base_url`
4. Set value to `https://reqres.in/api`
5. Save and select the environment

Importance

Environment configuration improves efficiency and ensures clean test design by avoiding hardcoded values.

5.3 API Key Configuration

Overview

API keys are typically used for authentication and authorization in secure APIs. They ensure that only authorized users can access the system.

In This Project (ReqRes API Context)

ReqRes is a **public demo API**, so:

- No API key is required
- No authentication token setup is needed for most endpoints
- Only the login API returns a sample token for learning purposes

Login Token Example

For the login API:

```
POST {{base_url}}/login
```

Response:

```
{
  "token": "QpwL5tke4Pnpja7X4"
}
```

How It Would Be Used (Conceptual Understanding)

In real-world APIs, API keys or tokens are used like:

- Passed in headers:

```
Authorization: Bearer <token>
```

Importance

Even though ReqRes does not require API keys, understanding this concept is important for real API testing scenarios in companies.

5.4 Newman Setup

Overview

Newman is a command-line tool used to run Postman collections outside the Postman application. It is used for **automation testing and CI/CD integration**.

Prerequisites

Before installing Newman:

- Node.js must be installed on the system
 - npm (Node Package Manager) should be available
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Installation Steps

Run the following command in terminal or command prompt:

```
npm install -g newman
```

This installs Newman globally on the system.

Verify Installation

To check if Newman is installed correctly:

```
newman -v
```

If installed successfully, it will show the version number.

Running a Collection

To execute Postman collection using Newman:

```
newman run ReqRes_Collection.json
```

Generating HTML Report

To generate a detailed test report:

```
newman run ReqRes_Collection.json -r html
```

Key Features Used

- Command-line execution of API tests
 - Bulk execution of test cases
 - HTML report generation
 - CI/CD pipeline compatibility
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Importance

Newman is important because:

- It automates API testing
 - It removes dependency on GUI (Postman UI)
 - It supports continuous testing in DevOps pipelines
 - It provides professional-level test execution reports
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5.5 Summary of Test Environment Setup

The test environment for this project is built using:

- **Postman** for manual API testing
- **Environment variables** for dynamic request handling
- **ReqRes API** as the test backend system
- **Node.js + Newman** for automation execution

This setup ensures a complete and scalable API testing environment that supports both manual and automated testing workflows.

5.6 Final Conclusion

A properly configured test environment is essential for successful API testing. In this project, Postman provides the testing interface, environment variables ensure flexibility, ReqRes API serves as the system under test, and Newman enables automation and reporting.

Together, these components create a professional-grade API testing setup similar to real-world QA industry standards.
